

ADITYA SINGH

Software Engineering Intern | Java Backend Developer

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SUMMARY

Computer Science student with strong foundations in Data Structures, Algorithms, and Object-Oriented Design. Experienced in Java backend engineering, utilizing modern Java features (Functional Interfaces, Stream API pipelines, Concurrency, and Generics) to develop structured architectures, generic components, and reliable, high-performance data layers. Currently learning Spring Boot and applying the core Spring Framework ecosystem (IoC, Dependency Injection, Annotations, Spring MVC) for modern web backend applications.

EDUCATION

B.Tech in Computer Science and Engineering 2023 – 2027 (Expected)
KIET Group of Institutions, Ghaziabad, Delhi-NCR, India CGPA: 7.00 / 10.0

TECHNICAL SKILLS

Advanced Java: Java Stream Pipelines, Functional Interfaces (Function, Predicate), Lambda Expressions, Generics, OOP Design, Multithreading & Concurrency (ConcurrentHashMap), Exception Handling, File I/O
Frameworks: Spring Framework (Core, IoC, DI, MVC, Annotations), Spring Boot (Learning), Hibernate / JPA
DSA & Languages: Data Structures & Algorithms (200+ solved problems on LeetCode), Java 8+, C++, Python, JavaScript (ES6+)
Web & Databases: SQL (MySQL/PostgreSQL), Node.js, Express, MongoDB, RESTful API design, HTML5, CSS3, React.js
Tools & Cloud: Git, GitHub, Maven, JUnit 5 (Unit Testing), VS Code, IntelliJ IDEA, AWS (Cloud Practitioner), Oracle Cloud

PROJECTS

JavaBank – Desktop Retail Banking Application *Java 17, JavaFX, Maven, JUnit 5*

- Developed a secure desktop banking system utilizing a clean **4-layer architecture** (presentation, service, repository, analytics) to separate concerns and improve maintainability.
- Designed a generic, thread-safe in-memory Repository pattern (`InMemoryRepository<T, ID>`) leveraging **Functional Interfaces** (`Function<T, ID>`, `Predicate<T>`) to allow dynamic ID extraction and generic data queries via lambda filters.
- Engineered custom checked exception handling with try-with-resources CSV persistence and registered a **JVM shutdown hook** to automatically serialize cached map configurations to files on sudden exit.
- Implemented optimized **Java Stream pipelines** utilizing primitive streams (`mapToDouble().sum()`, `average()`) to eliminate autoboxing overhead, and custom collectors (`groupingBy`, `counting`) to drive analytics dashboards.

Real-Time Bus Tracking & Management System *Node.js, Express, MongoDB, Socket.io, React.js (Leaflet, Recharts)*

- Engineered a real-time tracking application featuring secure role-based portals for transit operators, commuters, and system administrators.
- Integrated **Socket.io** to push live geographical locations, estimated arrival times (ETA), and bus crowd levels to active clients with low latency.
- Architected an enterprise-style 19-endpoint REST API using Node.js/Express, incorporating JWT and bcrypt-based authorization, ticket generation workflows, and MongoDB schemas.

SkinSavvy – Skincare Intelligence & Recommendation Platform *React.js, Node.js, Express, Local TF-IDF Engine*

- Built an intelligent full-stack recommendations system implementing a local, zero-external-API **RAG chatbot** using TF-IDF vectorization and cosine similarity across 113 product documents.
- Developed a conflict-detection engine handling ingredient incompatibilities (e.g., AHAs + Retinol) and matching user condition risk profiles (acne, eczema) against product formulas.
- Optimized search indexing routines to construct TF-IDF indexes within ~22ms at server startup, supporting instantaneous chatbot response times.

ACHIEVEMENTS

- **Smart India Hackathon (SIH) Finalist:** Led a developer team through problem definition, system architecture design, and final prototype presentation.

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- Oracle Cloud Infrastructure Certified AI Foundations Associate
- Palo Alto Networks Cybersecurity Foundation Course